Treadmill training remodels the muscle transcriptome

Abstract

We trained male mice for six weeks and profiled gastrocnemius by RNA-seq. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density.

1. Introduction

Earlier microarray work (GSE12345) examined acute exercise. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density.

2. Materials and methods

2.1. Animals and training

Male C57BL/6J mice (n = 6 per group) were assigned to sedentary control or treadmill training at 12 m/min for 60 min per day. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density.

Table 1. Sample groups used for RNA-seq.

Group	n	Sex	Tissue
Sedentary	6	M	Gastrocnemius
Trained	6	M	Gastrocnemius

2.2. Statistical analysis

Differential expression used DESeq2 with padj < 0.05 and |log2FC| > 1. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density.

3. Results

Training increased mitochondrial genes. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density.

Figure 1. Volcano plot of trained versus sedentary muscle.

4. Discussion

Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in gene expression, mitochondrial content and capillary density. Skeletal muscle adapts to repeated contraction through coordinated changes in

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Data availability

RNA-seq data are deposited in GEO under accession GSE209999.

References

1. Smith J. Exercise and muscle. Cell 2020.